SHIH-WEN LIU (CASPER)

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Research Interests

My core interests are **scalable collaborative intelligence**. I focus on designing AI systems that can actively collaborate with humans, other agents, and the physical world by identifying useful information, organizing knowledge over time, and supporting complex research workflows. Previously, I worked on cooperative robot learning, efficient multi-task learning, and multimodal large language models.

Education

- **National Cheng Kung University** — *Sep. 2024 – Present*
  Master of Science in Artificial Intelligence (Advisor: Prof. Wei-Ta Chu) — Tainan, Taiwan
  - CGPA: 4.00/4.00
- **National Sun Yat-sen University** — *Sep. 2020 – Jun. 2024*
  Bachelor of Science in Computer Science Engineering — Kaohsiung, Taiwan
  - CGPA: 3.62/4.00, Academic Excellence Award (2020)

Publications

[C.2] **Shih-Wen Liu**, Yen-Chang Chen, Wei-Ta Chu, Fu-En Yang, Yu-Chiang Frank Wang "**Frequency Switching Mechanism for Parameter-Efficient Multi-Task Learning**." In *Proceedings of the IEEE/CVF International Conference on Computer Vision (**CVPR**) 2026. (Work done in collaboration with NVIDIA.)*

[C.1] **Shih-Wen Liu**, Hsuan-Yu Fan, Wei-Ta Chu, Fu-En Yang, Yu-Chiang Frank Wang "**Histopathology Image Report Generation by Vision Language Models with Multimodal In-Context Learning**." In *the Conference of Medical Imaging with Deep Learning (**MIDL**) 2025. (Work done in collaboration with NVIDIA.)*

Projects

- **SUIII! A Ball Focus and Identity Memory for Robust Soccer Player Tracking** — *Feb. 2025*
  Tools: Python, PyTorch — [Project]
  - Improved soccer player tracking under occlusion using ball-focused reasoning, group motion dynamics, and long-term appearance memory
- **TRUMP: Task-Responsive Unified Multi-modal Perception** — *Sep. 2024*
  Tools: Python, PyTorch — [Project]
  - Developed a task-adaptive perception framework that extracts task-aware visual features from natural-language task descriptions
  - Designed a hypernetwork to generate additional task vectors for flexible task-specific adaptation
- **Split Conv: Efficient Convolution Layer Design** — *Aug. 2024*
  Tools: Pytorch — [Project]
  - Designed memory-efficient convolution and pooling layers for large-resolution training
  - Optimized implementations to reduce VRAM usage by 50 times
- **Jetson Nano Real-Time AR Teaching System** — *Jun. 2024*
  Tools: Python, OpenCV, Jetson — [Demo Video]
  - Developed AR overlay pipeline on Jetson Nano for live video processing
  - Achieved 30 FPS on 720p streams with real-time object detection
- **Desiary - A Novel Diary-Dating App** — *May 2024*
  Tools: Flask, React Native, Stable Diffusion, LoRA, MongoDB, Docker — [Demo Video]
  - Created diary-to-image generation using Stable Diffusion with LoRA fine-tuning
  - Built full-stack mobile app with user authentication and image gallery
  - Deployed backend in Docker with CI/CD pipelines for reliability
- **Newsbie - Your Little News Reporter** — *May 2024*
  Tools: Python, Flask, React Native, MongoDB, Docker — [Demo Video]
  - Implemented web-scraping modules to aggregate news from 10+ sources
  - Integrated LLM API for concise, on-demand article summarization and podcast generation pipeline
- **Let the Particles Have Babies! - PSO + GA + NN Mapping** — *Feb. 2024 - Jun. 2024*
  Tools: Python — [Project]
  - Combined Particle Swarm Optimization and Genetic Algorithms with neural networks for search-space reconstruction
- **Real-Time Multi-Filter App** — *Nov. 2022 - Dec. 2023*
  Tools: Python, NumPy — [Project]

- Developed 20+ stackable image filters (emboss, oil paint, pencil sketch) in pure NumPy
- Optimized vectorized operations to maintain 60+ FPS on 1080p frames

- **Glomerular Detection & Disease Classification** *Nov. 2022 - Dec. 2023*
  Tools: Python, PyTorch [Report Video]
  - Accelerated segmentation pipeline from 364 to 29 days via entropy-based filtering

- **Taiwanese Language Learning App Based on LLMs** *Sep. 2022 - Mar. 2024*
  Tools: Unity, Python, SpeechRecognition, TTS [Project]
  - Integrated Unity-based AR with an LLM-driven agent for immersive language-learning scenarios
  - Leveraged vision-language action commands to control interactive virtual agent
  - Developed real-time speech recognition and synthesis pipelines for pronunciation feedback

Experience

- **TAICA-Introduction to Artificial Intelligence []** *Fall 2025*
  Teaching Assistant Tainan, Taiwan
  - Selected course under the TAICA Program with students from 50+ universities.
  - Assisted in courses and assignments design.

- **NCKU CSIE-Linear Algebra []** *Spring 2024*
  Teaching Assistant Tainan, Taiwan
  - Assisted in courses and graded assignments.

- **NXP Semiconductors []** *Feb. 2023 – Aug. 2023*
  Industry Intern Kaohsiung, Taiwan
  - Developed automated data preprocessing tools, algorithms, and logic validation scripts for large-scale data.
  - Enhanced existing pipelines to improve performance and maintainability.

- **Scratch Programming Instructor** *Summer 2021 & 2022*
  Instructor Hsinchu, Taiwan
  - Taught introductory Scratch programming to elementary students, boosting communication and teaching skills.

Honors and Awards

- **Best Team Award** *2024*
  Critical Care Data Science Conference & Taiwan Datathon
  - Recognized for outstanding team collaboration and high-performance ICU data analysis with senior clinicians.

Skills

- **Programming Languages:** Python, C/C++, JavaScript, HTML, CSS
- **Data Science & Machine Learning:** NumPy, Pandas, PyTorch, TensorFlow, scikit-learn, Matplotlib
- **Developer Tools:** Docker, ROS2, Unity, React Native, Git, Bash
- **Specialized Areas:** Parameter-Efficient Fine-Tuning (LoRA), Adaptive Multi-Task Vision Models, Large Multi-Modal Model, App development